

Beyond Closed-Set Recognition: Geometric Open-Set Detection for Kenyan Sign Language

Rancy Chepchirchir

Abstract

Sign-language recognition systems are commonly evaluated under closed-set assumptions in which every input is presumed to belong to one of the classes observed during training. This assumption is problematic for deployment because a model may assign an unseen sign to a known vocabulary item with substantial confidence rather than indicate that the input is outside its learned vocabulary.

This work investigates this failure mode in a reconstructed four-class Kenyan Sign Language (KSL) Transformer baseline supporting the glosses *father*, *hello*, *is*, and *my*. We first reconstruct the checkpoint-compatible landmark representation and extract 256-dimensional Transformer embeddings. We then construct an embedding reference bank from 572 training samples and evaluate distance-based open-set detection on 124 held-out in-distribution samples and 450 natural out-of-distribution videos drawn from 30 KSL glosses outside the classifier vocabulary.

The closed-set classifier exhibits severe prediction collapse on the expanded unseen-vocabulary set: 423 of 450 videos (94.0%) are assigned to the class *my*, despite none belonging to the supported vocabulary. In contrast, nearest-neighbour distances in the learned embedding space separate the held-out in-distribution and evaluated natural-OOD samples. For mean 5-nearest-neighbour distance, the held-out ID mean is 1.1207 and the natural-OOD mean is 12.9992. The maximum observed held-out ID score is 7.4174, whereas the minimum observed natural-OOD score is 10.1010, yielding an empirical separation gap of 2.6836. Across $k \in \{1, 3, 5, 10, 20\}$, the evaluated dataset yields AUROC and AUPR of 1.0000 and FPR95 of 0.0000.

These results demonstrate a pronounced mismatch between closed-set confidence and vocabulary membership in this baseline, while showing that its learned representation contains substantial information for detecting the evaluated unseen KSL signs. The results should be interpreted as evidence on the present model and datasets rather than as proof of general open-set robustness.

1 Introduction

Automatic sign-language recognition has the potential to improve accessibility to communication technologies, yet recognition systems are frequently developed under a closed-set classification assumption. Under this formulation, a model receives an input sequence and must choose one label from a predefined vocabulary. Such a system has no explicit mechanism for stating that a valid sign lies outside its training vocabulary.

This limitation is particularly important when a classifier is deployed beyond the narrow conditions under which it was trained. A sign-language video may contain an unseen lexical item, a new signer, unusual landmark trajectories, incomplete detections, or other forms of distribution shift. Standard softmax classification nevertheless assigns probability mass across the known classes. Consequently, a high predicted probability need not imply that the input actually belongs to the model’s vocabulary.

In this work, we study this problem using a reconstructed Kenyan Sign Language baseline. The underlying classifier supports only four known glosses: *father*, *hello*, *is*, and *my*. Rather than

treating the classifier output as an adequate indicator of recognition reliability, we investigate the geometry of its learned 256-dimensional representation.

Our central question is whether an embedding learned for closed-set classification can also provide a useful signal for open-set rejection. We therefore construct a reference bank of training embeddings and score new inputs according to their distance from nearby training representations.

The study makes four principal contributions:

1. We reconstruct and document the preprocessing path required to obtain checkpoint-compatible KSL landmark sequences and Transformer embeddings.
2. We characterize the behaviour of the four-class classifier on natural KSL vocabulary outside its supported label set.
3. We evaluate nearest-neighbour embedding distance as an open-set score using held-out known-class samples and 450 natural unseen-vocabulary KSL videos spanning 30 glosses.
4. We expose the resulting open-set signal through the Lughalab inference API while retaining the original closed-set prediction for diagnostic transparency.

The objective is not to claim that a simple distance detector solves open-set KSL recognition generally. Instead, the experiments ask a narrower and reproducible question: *does the representation learned by this particular closed-set baseline contain sufficient geometric structure to distinguish its known vocabulary from the evaluated natural unseen vocabulary?*

2 System and Representation

2.1 Landmark representation

Each video is processed into a temporal sequence of landmark features. The native extracted representation contains 225 floating-point features per detected frame. For compatibility with the legacy checkpoint, the hand coordinates occupy the first 126 dimensions and the remaining 99 dimensions are zero padded before application of the checkpoint normalization statistics.

Let the extracted sequence be

$$X = [x_1, \dots, x_T]^\top \in \mathbb{R}^{T \times 225}.$$

The checkpoint expects a sequence of length 64. Sequences shorter than 64 frames are extended using repeat-last padding, whereas longer sequences are uniformly temporally downsampled. The resulting checkpoint-compatible sequence is

$$\tilde{X} \in \mathbb{R}^{64 \times 225}.$$

Checkpoint-specific feature-wise standardization is then applied using stored mean and standard-deviation statistics.

This representation was established through checkpoint reconstruction experiments. It should therefore be understood as a compatibility reconstruction and not as a claim that the unpublished original training preprocessing has been recovered exactly.

2.2 Transformer embedding

The reconstructed Transformer maps the normalized temporal sequence to a 256-dimensional sequence-level representation. We denote this mapping by

$$z = f_\theta(\tilde{X}), \quad z \in \mathbb{R}^{256},$$

LughaLab KSL inference and open-set recognition pipeline

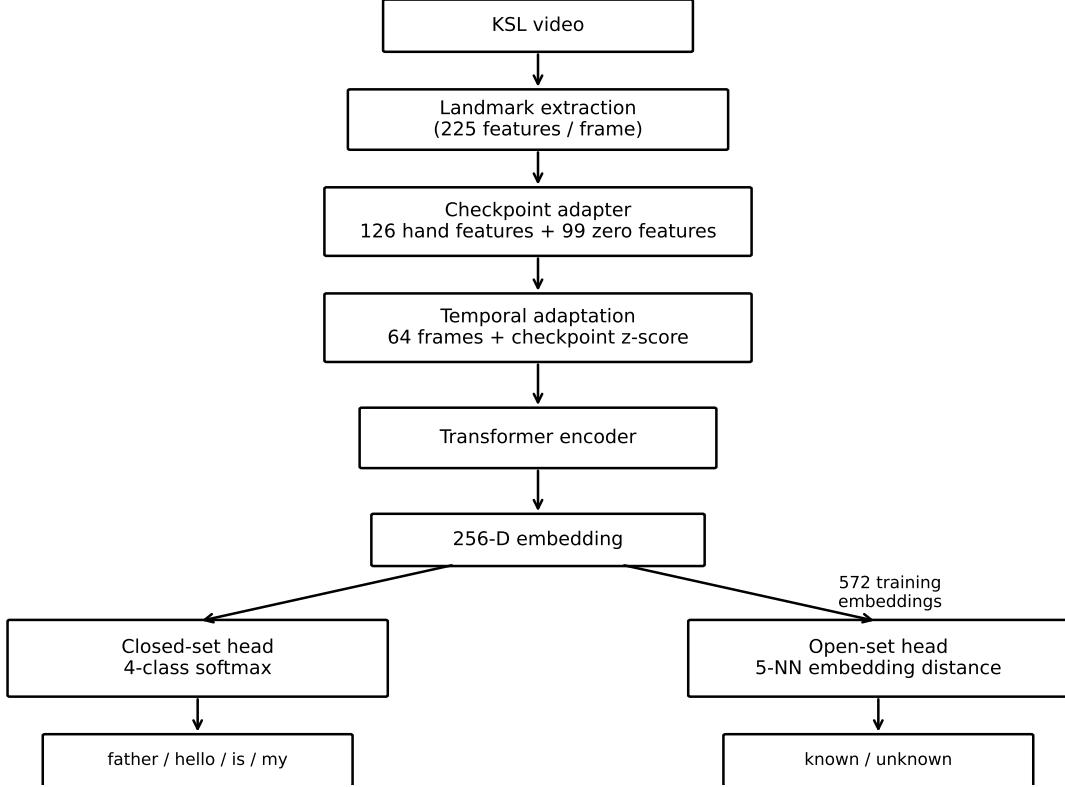


Figure 1: LughaLab KSL inference architecture. Video landmarks are converted into the reconstructed checkpoint representation and encoded as a 256-dimensional sequence embedding. The original closed-set branch produces a four-class prediction, while the proposed open-set branch measures distance to the training embedding reference bank and can reject an input as unknown.

where f_θ denotes the checkpointed encoder and z is its sequence representation. The classifier subsequently maps z to four logits

$$\ell = Wz + b,$$

followed by the standard softmax transformation

$$p(y = c \mid X) = \frac{\exp(\ell_c)}{\sum_{j=1}^4 \exp(\ell_j)}.$$

Because the softmax operates only over the four supported classes, it cannot by itself express membership in an unseen class.

3 Open-Set Detection

3.1 Embedding reference bank

A reference bank is constructed from 572 training samples:

$$\mathcal{B} = \{(z_i, y_i)\}_{i=1}^{572}.$$

For a query embedding z , Euclidean distances to all reference embeddings are computed:

$$d_i(z) = \|z - z_i\|_2.$$

Let $d_{(1)}(z), \dots, d_{(k)}(z)$ denote the k smallest distances. The open-set score is defined as the mean k -nearest-neighbour distance,

$$S_k(z) = \frac{1}{k} \sum_{j=1}^k d_{(j)}(z).$$

Larger values indicate that the query lies farther from the training representation bank and is therefore more geometrically novel.

3.2 Decision rule

Given threshold τ , the open-set decision is

$$\hat{o}(z) = \begin{cases} \text{unknown}, & S_k(z) > \tau, \\ \text{known}, & S_k(z) \leq \tau. \end{cases}$$

The runtime implementation uses $k = 5$ and a provisional threshold of

$$\tau_{0.99} = 5.4896,$$

obtained from the 99th percentile of the held-out in-distribution 5-NN score distribution.

The threshold is intentionally described as provisional: it represents an empirical operating point for the current baseline rather than a universally calibrated KSL open-set threshold.

4 Experimental Design

4.1 Known-class data

The embedding reference bank contains 572 training examples from the four supported classes. A separate held-out set of 124 known-class examples is used to characterize in-distribution distance behaviour.

The held-out samples are not added to the reference bank. This distinction is important because the open-set detector must be evaluated against known-class examples that were not used as reference points.

4.2 Natural unseen-vocabulary data

Natural OOD evaluation uses the eKitabu Kenyan Sign Language video dataset. The dataset contains 2,237 videos spanning 30 gloss categories. Its provided split metadata contains 1,487 training, 300 validation, and 450 test videos.

For the expanded natural-OOD experiment, all 450 videos from the provided test split are processed. There are 15 test videos per gloss and 30 evaluated glosses. None of these 30 gloss labels belongs to the four-class vocabulary of the baseline classifier.

All 450 videos were processed successfully. The mean landmark detection rate was 0.8091, with a minimum of 0.1955. The resulting natural-OOD embedding matrix has shape

$$450 \times 256.$$

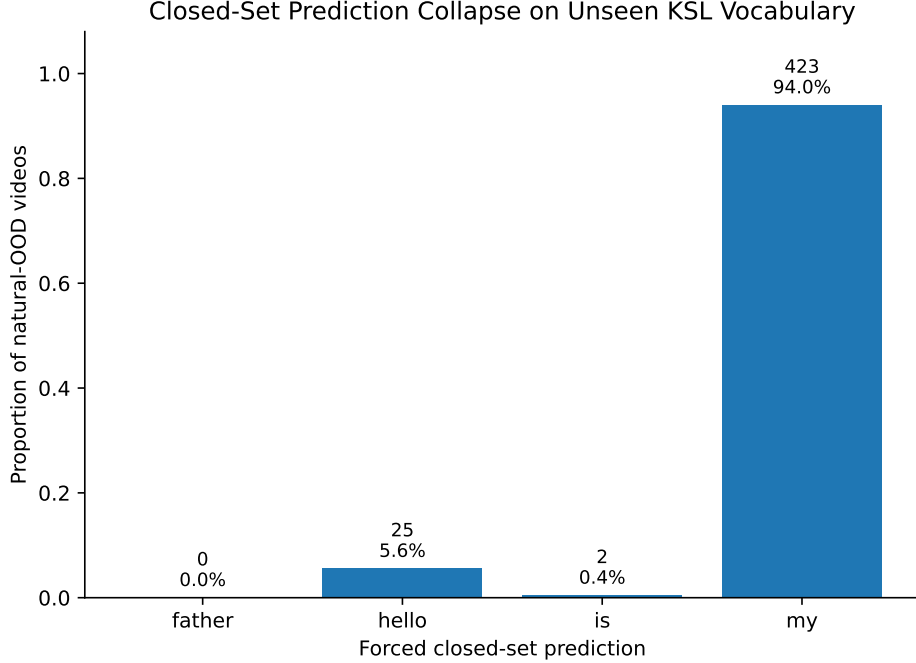


Figure 2: Forced closed-set predictions for 450 natural KSL videos whose glosses are absent from the four-class classifier vocabulary. The model assigns 94.0% of these videos to *my*.

4.3 Evaluation metrics

We evaluate k -NN distance detectors for

$$k \in \{1, 3, 5, 10, 20\}.$$

Reported measures include area under the receiver-operating characteristic curve (AUROC), area under the precision–recall curve (AUPR), false-positive rate at 95% true-positive rate (FPR95), empirical score distributions, and threshold-specific ID acceptance and OOD rejection rates.

We additionally examine the relationship between geometric novelty, maximum softmax probability, and landmark detection coverage.

5 Results

5.1 Closed-set behaviour on unseen vocabulary

The most immediate result is that the four-class classifier does not behave conservatively when presented with unseen KSL vocabulary. Of the 450 natural-OOD videos, 423 are classified as *my*, 25 as *hello*, and 2 as *is*. No video is assigned to *father*.

Thus,

$$\frac{423}{450} = 0.9400$$

of the entire unseen-vocabulary set collapses to a single known class.

The average maximum softmax probability is 0.6916, with values ranging from 0.3427 to 0.8348. Consequently, substantial closed-set confidence occurs even though every video in this experiment belongs to a gloss outside the supported vocabulary.

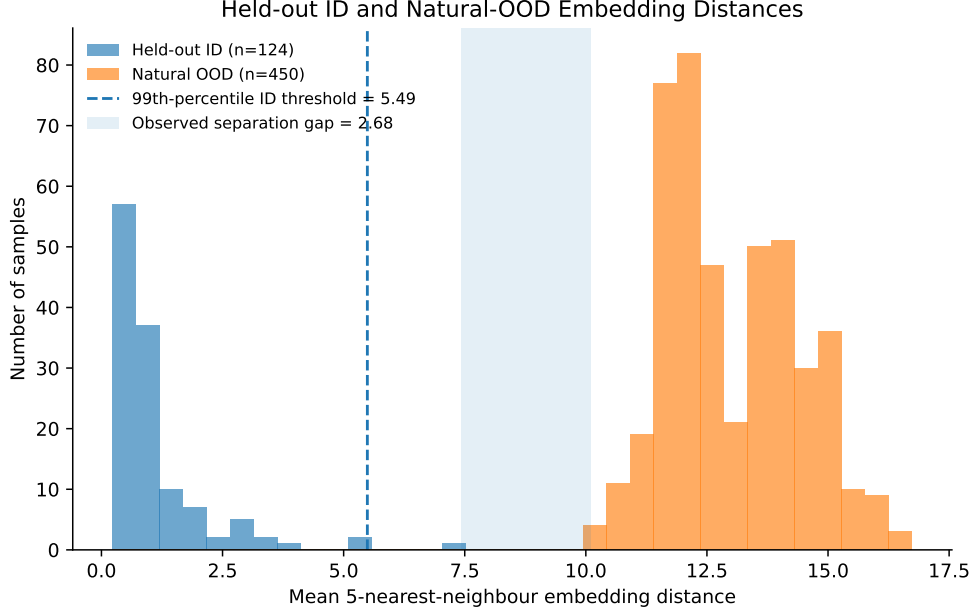


Figure 3: Distribution of mean 5-nearest-neighbour embedding distances for held-out known-class data and expanded natural unseen-vocabulary KSL data. The shaded region marks the observed separation between the largest held-out ID score and smallest natural-OOD score.

5.2 Embedding-space separation

The embedding-distance results differ sharply from the closed-set softmax behaviour.

For the 5-NN detector, held-out ID samples have mean score

$$\overline{S}_{5,\text{ID}} = 1.1207,$$

whereas natural-OOD samples have mean score

$$\overline{S}_{5,\text{OOD}} = 12.9992.$$

The largest observed held-out ID score is 7.4174 and the smallest natural-OOD score is 10.1010. Hence the evaluated samples exhibit an empirical gap of

$$10.1010 - 7.4174 = 2.6836.$$

5.3 Neighbourhood-size sensitivity

Table 1 reports performance for different values of k . All evaluated neighbourhood sizes produce complete ranking separation on the current dataset.

5.4 Threshold operating points

For $k = 5$, thresholds derived from held-out ID quantiles produce the operating characteristics shown in Table 2.

At the provisional 99th-percentile threshold, 98.39% of the held-out ID examples are accepted and all 450 evaluated natural-OOD examples are rejected.

Table 1: Open-set detection performance across neighbourhood sizes. Values describe the present held-out ID and natural-OOD evaluation sets.

k	AUROC	AUPR	FPR95	ID mean	OOD mean
1	1.0000	1.0000	0.0000	0.8382	11.6127
3	1.0000	1.0000	0.0000	0.9996	12.5170
5	1.0000	1.0000	0.0000	1.1207	12.9992
10	1.0000	1.0000	0.0000	1.3858	13.7757
20	1.0000	1.0000	0.0000	1.7720	14.7601

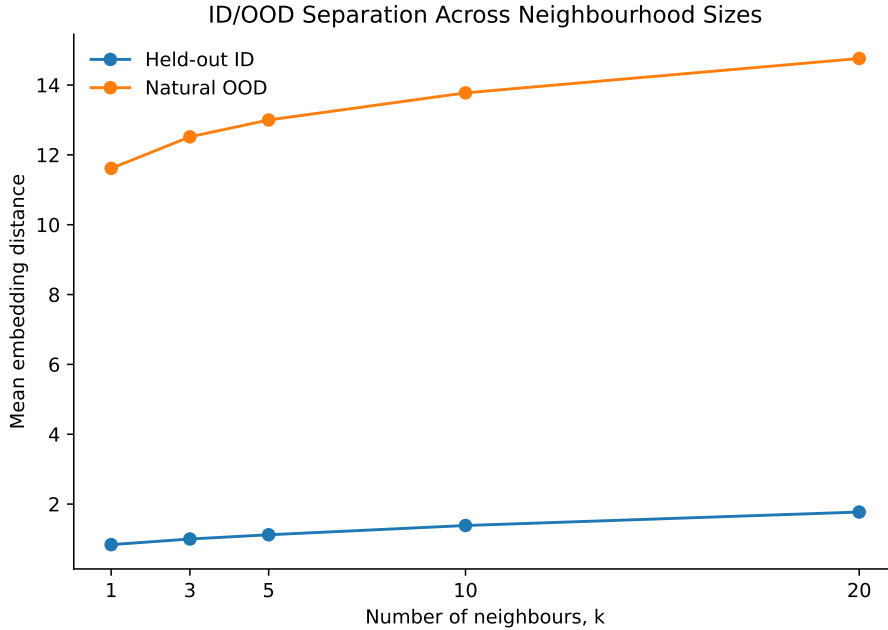


Figure 4: Mean ID and natural-OOD embedding distances as the number of neighbours is varied.

5.5 Variation across unseen glosses

The separation is not produced by only a small number of particularly distant unseen categories. At the 99th-percentile threshold, every evaluated gloss has an OOD rejection rate of 1.0000.

Mean 5-NN scores vary across glosses. For example, *agreement* has a mean score of 11.8429, *teach* 12.1715, *market* 14.1230, and *ugali* 13.7618. Even the smallest individual natural-OOD score in the expanded experiment, 10.1010, remains above the largest observed held-out ID score.

5.6 Softmax confidence and geometric novelty

Maximum softmax probability is negatively correlated with the 5-NN novelty score:

$$r = -0.6346.$$

The relationship demonstrates that the classifier can become less confident as inputs move farther from the training representation, but this does not make softmax confidence an explicit open-set detector. Natural-OOD inputs still receive ordinary known-class predictions, including maximum softmax probabilities as high as 0.8348.

Table 2: Empirical 5-NN operating points obtained from held-out ID quantiles.

ID quantile (%)	Threshold	ID accept	ID reject	OOD reject
90.0	2.3673	0.8952	0.1048	1.0000
95.0	3.1067	0.9435	0.0565	1.0000
97.5	3.8914	0.9677	0.0323	1.0000
99.0	5.4896	0.9839	0.0161	1.0000

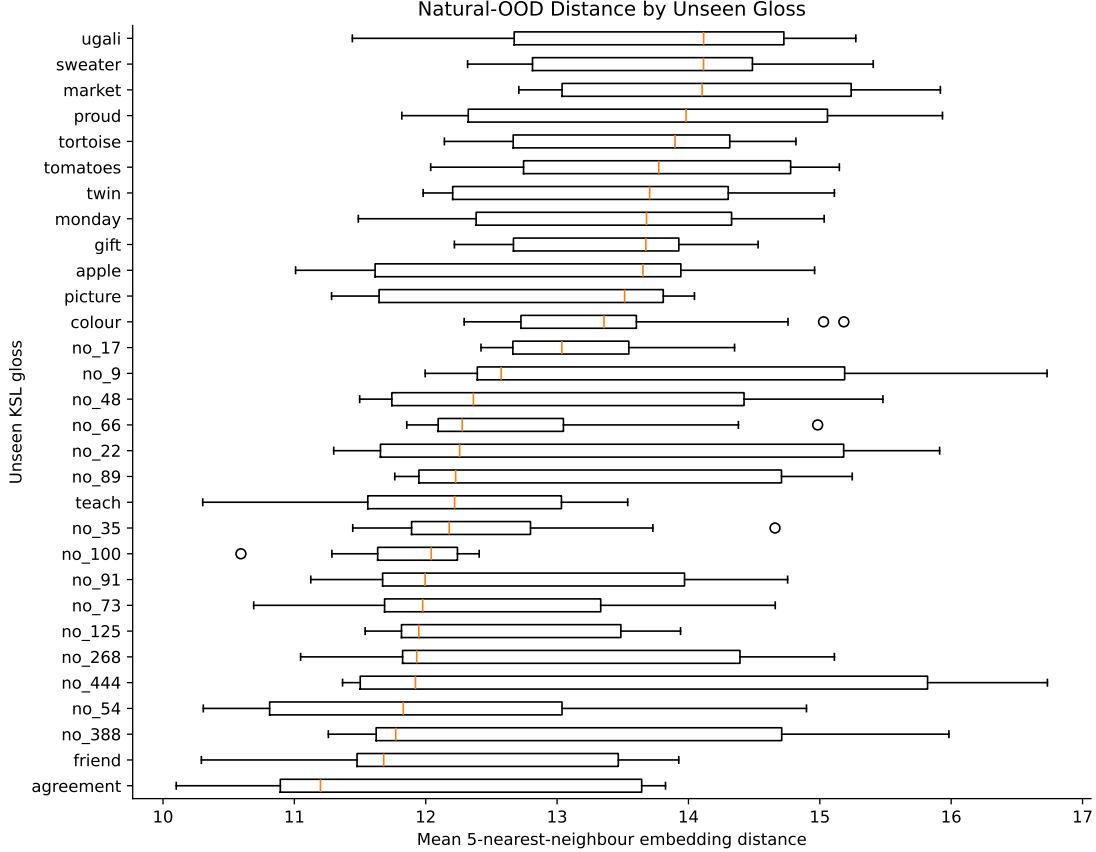


Figure 5: Distribution of 5-NN embedding novelty scores across the 30 unseen KSL glosses in the expanded natural-OOD evaluation.

5.7 Landmark coverage

Landmark detection coverage varies substantially across the expanded natural-OOD set. Mean detection rate is 0.8091, median detection rate is 0.9851, and the observed range is 0.1955–1.0000.

Coverage and 5-NN novelty are negatively correlated,

$$r = -0.5842.$$

This association is an important potential confound. The experiment therefore does not establish that embedding distance responds exclusively to lexical novelty; part of the distance variation may reflect landmark quality or other dataset differences.

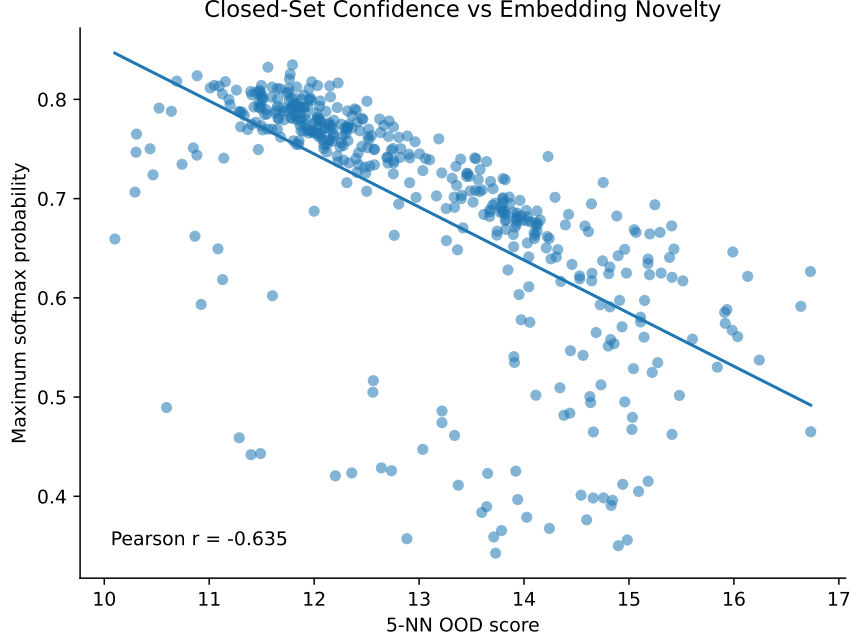


Figure 6: Maximum closed-set softmax probability versus 5-NN geometric novelty for the 450 natural-OOD videos.

6 Discussion

The experiments reveal two distinct behaviours of the same model. At the classifier output, unseen vocabulary is forced into the four known categories. At the representation level, however, the same inputs occupy regions that are substantially farther from the training embedding bank than held-out known-class inputs.

This distinction matters because softmax confidence answers a conditional closed-set question: assuming the input must belong to one of the supported classes, which class receives the largest probability? It does not directly answer whether that assumption is valid. Embedding distance provides a separate signal addressing that second question.

The 94% collapse of unseen vocabulary to *my* is also not merely a low-confidence ambiguity phenomenon. The natural-OOD set has mean maximum softmax probability 0.6916, and some unseen signs receive probabilities above 0.8. A production interface displaying these predictions without an explicit vocabulary-membership check could therefore present unsupported classifications with apparently meaningful confidence.

The geometric results suggest that useful open-set structure can exist even when the original model was trained for closed-set classification. This is attractive in a low-resource setting because it permits an open-set layer to be investigated without immediately retraining the underlying network.

Nevertheless, the observed perfect ranking metrics should not be overinterpreted. AUROC and AUPR of 1.0 describe the finite datasets evaluated here. They do not imply that future unknown signs, new recording environments, new signers, or adversarially selected inputs will always be separated from known classes.

7 Limitations

Several limitations constrain the conclusions of this study.

First, the known classifier vocabulary contains only four glosses. This makes the baseline particularly narrow and may increase the geometric separation between known and unseen

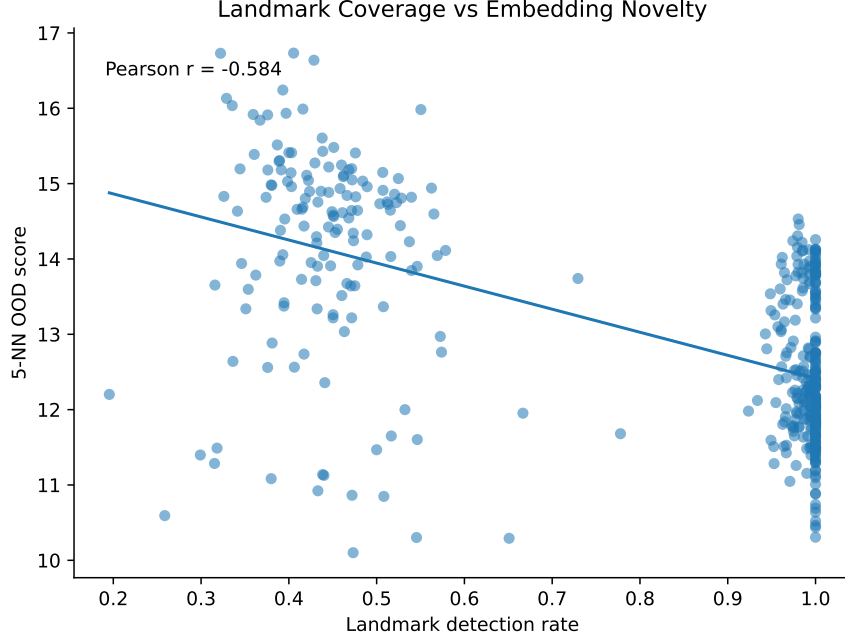


Figure 7: Landmark detection coverage versus 5-NN embedding novelty on the expanded natural-OOD set.

vocabulary. Open-set behaviour should be re-evaluated as the known vocabulary grows.

Second, the natural-OOD videos originate from an external dataset and may differ from the known-class data in ways other than lexical identity. Differences in signer population, camera characteristics, framing, motion, recording conditions, or landmark quality could contribute to embedding separation.

Third, landmark coverage is correlated with the OOD score. Although natural-OOD separation remains large in this experiment, future work should explicitly disentangle lexical novelty from extraction quality.

Fourth, the current threshold is based on empirical held-out ID quantiles. A deployment threshold should be selected using a dedicated calibration protocol and independently evaluated on data not involved in threshold choice.

Finally, the checkpoint preprocessing was reconstructed for compatibility. Although the resulting pipeline has been audited experimentally, it is not claimed to reproduce unpublished original training code exactly.

8 Conclusion

This work examines open-set failure in a four-class Kenyan Sign Language Transformer baseline. The classifier is shown to assign natural unseen-vocabulary KSL videos to known classes with substantial confidence, with 94.0% of 450 external test videos collapsing to the class *my*.

The model’s 256-dimensional representation tells a different story. Mean nearest-neighbour distance to a training embedding bank strongly separates the evaluated held-out known-class samples from the 30-gloss natural unseen-vocabulary set. For $k = 5$, the maximum observed held-out ID score is 7.4174 and the minimum natural-OOD score is 10.1010. Across all tested neighbourhood sizes, the present dataset yields AUROC and AUPR of 1.0000.

The central implication is therefore not that open-set KSL recognition is solved, but that closed-set probability and vocabulary membership are materially different quantities. For this baseline, explicitly modelling the latter through representation geometry provides a substantially

more appropriate mechanism for rejecting the evaluated unseen signs.